10TH CLASS CHEMISTRY — NEW SYLLABUS 2025-26 (PECTAA)

Chapter 14 — States of Matter and Phase Changes

Solved Quick Checks + Exercise | New Book (PECTAA 2025-26)

Quick Checks (Solved)

Quick Check 14.1

1. What is translational motion?

Translational motion is the movement of a particle from one place to another in a straight line. Gas particles move freely in all directions and liquid particles also slide over one another, so both show translational motion. Solid particles only vibrate about their fixed positions and cannot move from place to place, so solids do not show translational motion.

2. Is the average kinetic energy possessed by the particles of a gas and a liquid same at the same temperature?

Yes, it is the same. According to the kinetic particle theory, the average kinetic energy of particles depends only on the temperature and not on the physical state. At the same temperature, the average kinetic energy of the particles of a gas is equal to that of the particles of a liquid.

Quick Check 14.2

1. Why does water vapour at 100 °C cause more severe burns as compared to liquid water at the same temperature?

Water vapour (steam) at 100 °C carries extra hidden heat, which is called latent heat of vaporisation. This heat was taken during boiling to separate the molecules and change liquid water into vapour. When steam touches the skin, it condenses back into water at 100 °C and gives out this hidden heat to the skin. Boiling water at 100 °C has no such extra heat to release, so steam causes more severe burns.

2. Why do we feel comfortable wearing cotton clothes in summer?

Cotton is a good absorber of moisture and it allows air to pass through it. In summer the sweat produced by the body is absorbed by cotton and then evaporates. Evaporation takes heat from the body, so a cooling effect is produced. Cotton also lets air circulate, and for these two reasons we feel comfortable in cotton clothes while clothes made of synthetic fibre feel hot and sticky.

Quick Check 14.3

1. Does the change in temperature occur during the process of evaporation?

Yes, a small change occurs. During evaporation the fast moving molecules, which have above-average kinetic energy, leave the surface of the liquid. The molecules left behind have less average kinetic energy, so the remaining liquid becomes slightly cool. That is why evaporation produces a cooling effect, for example water in an earthen surahi becomes cold.

2. Does ice sublime?

Yes, ice can sublime, but very slowly. Below 0 °C the molecules of ice may pass directly into water vapour without melting first. This is why ice cubes kept in a freezer for a long time become smaller and the frost on frozen food slowly disappears (this is also used in freeze drying of food).

3. Give one example each for sublimation and deposition other than mentioned in the text.

Sublimation: iodine crystals give violet vapours on gentle heating; camphor and ammonium chloride also sublime. **Deposition:** white frost forms directly on the inner side of window panes in winter; iodine vapour changes directly into solid crystals on a cold watch glass.

Quick Check 14.4

(i) With the help of kinetic particle theory explain how a gas exerts pressure.

Gas particles are in a continuous state of random motion. They move with high speed and collide with the walls of the container again and again. Each collision applies a small force on the wall, and the total force of millions of such collisions per unit area of the wall is the pressure of the gas. When the temperature is raised, the particles move faster, the collisions become more frequent and harder, so the pressure of the gas increases (at constant volume).

(ii) How does the volume of a gas change when its temperature is decreased? (Pressure and mass constant)

The volume of the gas decreases. On cooling, the average kinetic energy and speed of the particles decrease, so they collide with the walls with less force. To keep the pressure constant the gas must occupy less space. This is Charles' law: at constant pressure the volume of a fixed mass of gas is directly proportional to its absolute temperature ($V \propto T$), so volume falls when temperature falls.

Quick Check 14.5

1. Which gas among NO_2, SO_2 and H_2S will diffuse at the fastest speed and which at the slowest speed under similar conditions?

Molar masses are NO_2 = 46 g mol^{-1}, SO_2 = 64 g mol^{-1} and H_2S = 34 g mol^{-1}. At the same temperature all gases have the same average kinetic energy, so the lighter gas moves faster. **H_2S diffuses fastest** (lightest) and **SO_2 diffuses slowest** (heaviest). The rate of diffusion is inversely proportional to the square root of molar mass.

2. Explain with the help of kinetic particle theory why ammonia gas diffuses faster than hydrogen chloride gas.

At the same temperature, the molecules of all gases possess the same average kinetic energy. The mass of an ammonia molecule (17 g mol^{-1}) is less than that of a hydrogen chloride molecule (36.5 g mol^{-1}). The lighter ammonia molecules therefore move faster and cover a larger distance in the same time. In the glass tube experiment, ammonia travelled 59.5 cm while HCl travelled only 40.5 cm in the same duration, which proves that ammonia diffuses faster.

Exercise (Solved)

A. Multiple Choice Questions

1. According to kinetic theory, the basic difference between solid, liquid and gas is due to:

(a) the difference in the movements of the particles.
(b) the chemical properties of the particles.
(c) the size of the particles.
(d) the shape of the particles.

Answer: (a) — All three states are chemically the same; only the motion, arrangement and distance between particles are different.

2. Upon heating the rate of evaporation:

(a) decreases
(b) increases
(c) remains the same
(d) initially decreases and then increases upon further heating

Answer: (b) — Heating increases the kinetic energy, so more molecules escape from the surface in the same time.

3. The inter-particle attractions are the strongest in:

(a) Solids
(b) Liquids
(c) Plasma
(d) Gas

Answer: (a) — In solids the particles are closest and held in fixed positions, so the attractive forces are strongest.

4. Cooling the vapours of some gases changes them directly into solid state. This phenomenon is called:

(a) evaporation
(b) condensation
(c) sublimation
(d) deposition

Answer: (d) — Deposition is the direct change of a gas into a solid, for example formation of frost in winter.

5. Physical state in which the particles possess the maximum energy:

(a) Solid
(b) Liquid
(c) Gaseous
(d) Vapour

Answer: (c) — Gas particles have the highest kinetic energy because they move fastest and are farthest apart.

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6. How does the process of evaporation depend on the force of attraction among the molecules of a liquid?

(a) It decreases with the increasing strength of attraction.
(b) It increases with the increasing strength of attraction.
(c) It is independent of the strength of attraction.
(d) It first increases and then decreases with the increasing strength of the force of attraction.

Answer: (a) — A strong force of attraction holds the molecules at the surface, so fewer molecules can escape.

7. Which gas will diffuse at the fastest rate?

(a) HCl
(b) SO_2
(c) H_2S
(d) CO_2

Answer: (c) — H_2S has the smallest molar mass (34 g mol^{-1}), so its molecules move fastest.

8. The phase changes A and B in gas → A → liquid → B → solid are:

(a) Melting, evaporating
(b) Condensation, melting
(c) Condensation, freezing
(d) Boiling, freezing

Answer: (c) — Gas to liquid is condensation and liquid to solid is freezing.

9. A student noticed frost falling on the ground at 6:30 am and disappearing by 8:30 am. Which phase change was responsible for changing the frost into water vapours?

(a) Melting
(b) Evaporation
(c) Sublimation
(d) Deposition

Answer: (c) — The solid frost changed directly into vapours without melting, which is sublimation.

B. Short Answer Questions

14.1 From where does the energy come when a liquid evaporates?

The energy comes from the liquid itself and also from the surroundings. The molecules present at the surface which already possess above-average kinetic energy escape into the air. As these fast molecules leave, the average energy of the remaining liquid falls and it becomes cooler. It then absorbs heat from the surroundings, and this heat keeps the evaporation going. No external source of heat is needed.

14.2 Is condensation an endothermic process?

No, condensation is an exothermic process. When a gas is cooled, its molecules lose kinetic energy, come closer and are converted into a liquid. In this process the molecules give out heat to the surroundings, for example steam condensing on a cold glass releases heat and the glass becomes warm.

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14.3 Why do naphthalene balls (used to repel insects) disappear after some time?

Naphthalene balls sublime. At room temperature the solid naphthalene changes directly into vapours without melting, so no liquid is left behind. The vapours spread in the air and their strong smell keeps the insects away from clothes. With the continuous loss of vapours the balls become smaller and after some time they disappear completely.

14.4 Why does the temperature remain constant during a phase change?

During a phase change the heat supplied is not used to increase the kinetic energy of the particles; it is used as latent heat to overcome the forces of attraction between the particles. As the average kinetic energy does not change, the temperature also does not change until the whole substance has changed its state. For example, ice remains at 0 °C until all of it melts into water.

14.5 Is it possible to compress a liquid like a gas?

No, a liquid cannot be compressed like a gas. In a liquid the particles are already very close to one another and there is very little empty space between them, so there is almost no space left to press them further. A gas can be compressed easily because its particles are far apart with large empty spaces between them.

14.6 Can you change the temperature at which water boils in an experiment in your lab?

Yes, the boiling point of water can be changed by changing the external pressure. A liquid boils when its vapour pressure becomes equal to the external pressure. If the external pressure is reduced, for example by a vacuum pump, water boils below 100 °C. If the external pressure is increased, as in a pressure cooker, water boils above 100 °C.

14.7 In which season do wet clothes get dry after a relatively longer time?

In winter and in the rainy season wet clothes dry after a longer time. In winter the temperature is low, so the molecules of water have less kinetic energy and evaporate slowly. In the rainy season the air is already full of water vapour, so it can take very little more vapour and evaporation becomes slow. In hot and dry summer the clothes dry quickly.

14.8 When a solid is heated, what happens to its particles which are vibrating?

On heating, the particles of the solid gain kinetic energy and vibrate faster about their mean positions. The amplitude of vibration increases and the forces of attraction between the particles become weaker. At the melting point the vibrations become so strong that the particles leave their fixed positions and start sliding over one another, so the solid melts into a liquid.

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14.9 Do solids and liquids also diffuse just like gases?

Yes, solids and liquids also diffuse, but very slowly as compared to gases. In liquids the particles are close together and strongly attracted, so they can only slip past their neighbours; a drop of ink spreads slowly in water. In solids the particles have fixed positions and only vibrate, so diffusion is extremely slow; two polished metal blocks pressed together for a long time show a thin layer of one metal inside the other.

14.10 Why is dew formed in the early hours of the morning?

At night the ground and the air near it lose heat and become cold. Cold air can hold less water vapour than warm air, so the vapour present in the air becomes excess. This extra vapour condenses on the cool surfaces of leaves, grass and stones as tiny water droplets. These droplets are called dew, and they are seen in the early hours of the morning.

C. Constructed Response Questions

14.1 Differentiate between evaporation and boiling.

Evaporation	Boiling
It is a surface phenomenon; it occurs only from the surface of the liquid.	It is a bulk phenomenon; it occurs throughout the whole liquid and bubbles are formed inside it.
It occurs slowly at all temperatures, but below the boiling point.	It occurs quickly at a specific temperature called the boiling point, at a fixed external pressure.
It produces cooling, because the fast molecules leave the liquid.	It does not produce cooling, because heat is supplied continuously from outside.
It needs a small amount of energy, which is provided from inside the liquid and from the surroundings.	It needs a constant supply of energy from an external source, such as a burner.
The temperature of the liquid does not remain constant; it falls slightly.	The temperature of the liquid remains constant until the whole liquid changes into vapour.

14.2 Describe how boiling is related to the external pressure.

A liquid boils when the vapour pressure of the liquid becomes equal to the external pressure acting on its surface. If the external pressure is low, less heat is needed to make the vapour pressure equal to it, so the liquid boils at a lower temperature. That is why water boils at about 98 °C in Murree, where the atmospheric pressure is low, but at 100 °C in Karachi, where the pressure is normal. If the external pressure is high, as inside a pressure cooker, the liquid boils at a higher temperature. Hence the boiling point of a liquid increases with the increase of external pressure and decreases with the decrease of external pressure.

14.3 Why the process of evaporation does not need the input of energy?

Evaporation does not need any external supply of energy because only those molecules escape from the surface which already possess more than the average kinetic energy. This energy is already present inside the liquid. When these fast molecules leave, the remaining liquid becomes cooler and takes heat from the surroundings, and this heat keeps the process going. Boiling, on the other hand, needs a continuous external supply of heat because the whole liquid has to be converted into vapour.

14.4 Why the heating and cooling curves for the phase changes adopt the same path?

A phase change is a reversible process and it takes place at the same temperature in both directions. Ice melts at 0 °C and water freezes at 0 °C; water boils at 100 °C and steam condenses at 100 °C. The amount of latent heat absorbed during melting or boiling is exactly the same as the amount released during freezing or condensation. Therefore the cooling curve from F to A is just the reverse of the heating curve from A to F; only the direction of heat flow is different, not the path.

14.5 Why ice melts when the pressure is exerted on it?

Ice has an open cage-like structure with empty spaces between its molecules, and its density is less than that of water. When pressure is applied, the molecules are pushed into these empty spaces and the open structure breaks. This lowers the melting point of ice, so a thin layer of ice melts into water even below 0 °C. When the pressure is released, the water freezes again and joins the pieces together. This is why a snowball can be made by pressing snow in the hands.

14.6 Why do phase changes occur?

Phase changes occur because of a change in the internal energy of the particles of a substance. When heat is supplied, the kinetic energy of the particles increases, they move or vibrate faster and the forces of attraction between them become weak; as a result a solid changes into a liquid and a liquid into a gas. When heat is removed, the kinetic energy of the particles decreases, they come closer and the attractive forces become strong; as a result a gas changes into a liquid and a liquid into a solid. Hence every phase change is caused by a gain or loss of energy, that is, by heating or cooling.

D. Descriptive Questions

14.1 Explain the difference in the rates of diffusion of two gases based on kinetic theory.

According to the kinetic particle theory, the molecules of all gases possess the same average kinetic energy at the same temperature. The kinetic energy of a molecule depends upon its mass and speed (KE = ½ mv^2). Therefore the gas with lighter molecules must move with a greater speed and the gas with heavier molecules with a smaller speed. This is why hydrogen (2 g mol^{-1}) diffuses much faster than oxygen (32 g mol^{-1}).

This can be proved by the glass tube experiment. Cotton plugs soaked in ammonia solution and hydrochloric acid are placed at the two ends of a 100 cm long glass

tube at the same time. White fumes of ammonium chloride are formed where the two gases meet. Ammonia covers 59.5 cm while HCl covers only 40.5 cm in the same time, which shows that the lighter gas diffuses faster. The rate of diffusion is inversely proportional to the square root of the molar mass, so the ratio of the rates of diffusion of ammonia and HCl is √(36.5/17) = 1.47, which is equal to 59.5/40.5.

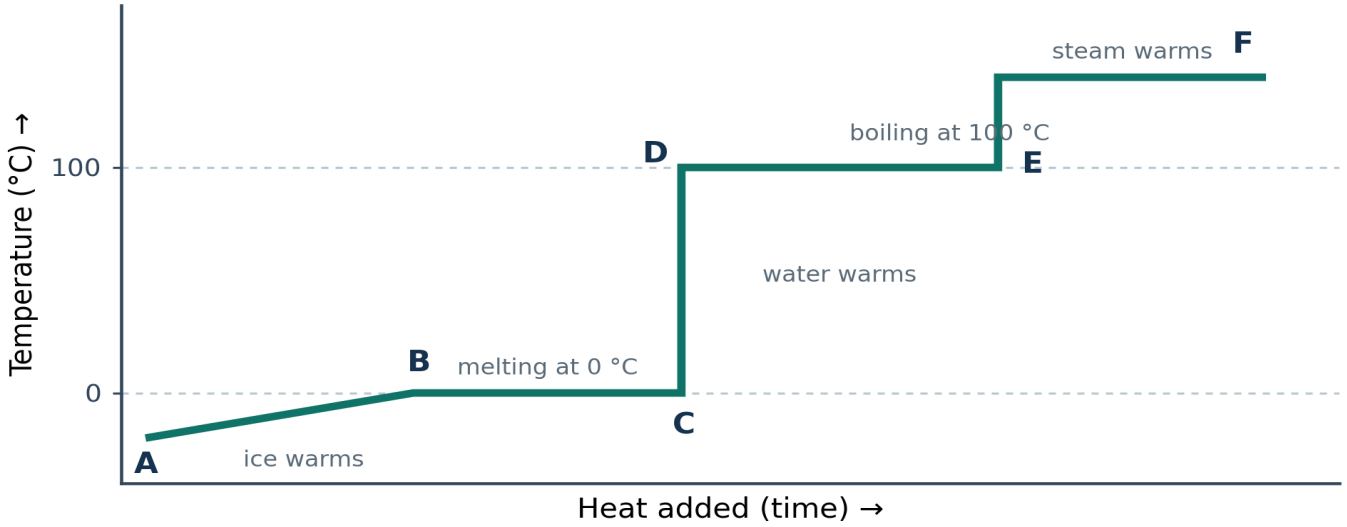
Temperature also affects diffusion: at a higher temperature the molecules have more kinetic energy, so they move and spread faster.

14.2 How does the process of sublimation helpful in printing and in the usage of air freshner?

(a) Solid air fresheners: A solid air freshener contains a scented substance which sublimes slowly at room temperature. When it is exposed to the atmosphere, its solid particles gain enough energy to overcome the attractive forces holding them together, and they spread in the room as sweet-smelling vapours. These vapours mask unpleasant odours. The solid keeps shrinking as it sublimes and after some time it disappears completely, without leaving any liquid behind.

(b) Sublimation printing: In sublimation printing the design is first printed on a special paper with sublimation inks. This paper is placed on the fabric and heat and pressure are applied. The heat opens the pores of the fabric and the ink changes into vapour. The vapour enters the pores of the fabric, cools down and changes back into a solid inside the fabric. As the ink is embedded inside the fabric and not just attached to its surface, the printing is permanent and does not fade with time. This method is used for printing on T-shirts, mugs, ceramic, wood and metal articles.

14.3 Draw heating curve for the physical changes water undergoes with change in temperature.



Temperature is taken along the vertical axis and heat added (time) along the horizontal axis. **A to B:** ice is heated and its temperature rises from below 0 °C to 0 °C; the kinetic energy of the particles increases but the state does not change.

B to C: At 0 °C the curve becomes flat and the temperature remains constant. The heat supplied here is used as latent heat to break the forces between the molecules of ice, so the ice melts into water. This temperature is the melting point.

C to D: The temperature of water rises again as it is heated, because the average kinetic energy of the water molecules increases.

D to E: At 100 °C the curve again becomes flat. The heat supplied is now used to change water into steam without raising the temperature. This temperature is the boiling point.

E to F: steam is heated further and its temperature rises above 100 °C. If the same graph is read backwards from F to A, it becomes the cooling curve, in which steam condenses at 100 °C and water freezes at 0 °C.

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